

Exercise J01

Q1. If $\sum_{i=1}^{20} \sin^{-1} x_i = 10\pi$ then $\sum_{i=1}^{20} x_i = 10\pi$ is equal to -

- (A) 20 (B) 10 (C) 0 (D) None of these

Ans. A

Sol. -

Q2. If $\cos^{-1}(a) + \cos^{-1}(b) + \cos^{-1}(c) = 3\pi$ and $f(1) = 2$, $f(x+y) = f(x)f(y)$ for all x, y ; then $a^{2f(1)} + b^{2f(2)} + c^{2f(3)} + \frac{(a+b+c)}{a^{2f(1)}b^{2f(2)}c^{2f(3)}}$ is equal to -

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. -

Q3. If $\sin^{-1}x + \tan^{-1}x = y$ ($-1 < x < 1$), then which is not possible -

- (A) $y = \frac{3\pi}{2}$ (B) $y = 0$ (C) $y = \frac{\pi}{2}$ (D) $y = -\frac{\pi}{2}$

Ans. A

Sol. -

Q4. The value of $\sin^2\left(\cos^{-1}\frac{1}{2}\right) + \cos^2\left(\sin^{-1}\frac{1}{3}\right)$ is-

- (A) $\frac{17}{36}$ (B) $\frac{59}{36}$ (C) $\frac{36}{59}$ (D) None of these

Ans. B

Sol. -

Q5. $\cot\left[\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8}\right] =$

- (A) 1 (B) -1 (C) $\sqrt{2}$ (D) $-\sqrt{2}$

Ans. A

Sol. -

Q6. $\cot[\tan^{-1}(1/7) + \tan^{-1}(1/13)]$ is equal to -

- (A) 2/9 (B) 9/2 (C) 7/9 (D) 9/7

Ans. B

Sol. -

Q7. If $0 < x < 1$, then $\sqrt{1+x^2} [\{x \cos(\cot^{-1} x) + \sin(\cot^{-1} x)\}^2 - 1]^{1/2} =$

(A) $\frac{x}{\sqrt{1+x^2}}$ (B) x (C) $x\sqrt{1+x^2}$ (D) $\sqrt{1+x^2}$

Ans. C

Sol. $\sqrt{1+x^2} [\{x \cos(\cot^{-1} x) + \sin(\cot^{-1} x)\}^2 - 1]^{1/2}$

$$\sqrt{1+x^2} \left[\left\{ x \frac{x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}} \right\}^2 - 1 \right]^{1/2}$$

$$\sqrt{1+x^2} \left[\frac{(x^2+1)^2}{x^2+1} - 1 \right]^{1/2}$$

$$\sqrt{1+x^2} (x)$$

Q8. If $a \leq \tan^{-1}x + \cot^{-1}x + \sin^{-1}x \leq b$. Then-

(A) $a = 0, b = \pi$ (B) $b = \frac{\pi}{2}$ (C) $a = \frac{\pi}{4}$ (D) None of these

Ans. A

Sol. -

Q9. $\sin^{-1}x + \sin^{-1} \frac{1}{x} + \cos^{-1}x + \cos^{-1} \frac{1}{x} =$

(A) π (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{2}$ (D) None of these

Ans. A

Sol. -

Q10. If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$, then $\cos^{-1}x + \cos^{-1}y =$

(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) π

Ans. B

Sol. $\sin^{-1}x + \sin^{-1}y = \frac{\pi}{2} - \cos^{-1}x - \cos^{-1}y + \frac{\pi}{2} = \pi - (\cos^{-1}x + \cos^{-1}y) = \frac{2\pi}{3}$

$$\Rightarrow \cos^{-1}x + \cos^{-1}y = \frac{\pi}{3}$$

Q11. $\sin \left[2 \cos^{-1} \left(-\frac{3}{5} \right) \right]$ is equal to -

(A) $\frac{6}{25}$ (B) $\frac{24}{25}$ (C) $\frac{4}{5}$ (D) $-\frac{24}{25}$

Ans. D

Sol. -

Q12. If $\cos^{-1} \frac{3}{5} - \sin^{-1} \frac{4}{5} = \cos^{-1} x$, then $x =$

- (A) 0 (B) 1 (C) $1/2$ (D) $1/4$

Ans. B

Sol. -

Q13. If $\sin^{-1} \frac{1}{3} + \sin^{-1} \frac{2}{3} = \sin^{-1} x$, then x is equal to -

- (A) 0 (B) $\frac{\sqrt{5} - 4\sqrt{2}}{9}$ (C) $\frac{\sqrt{5} + 4\sqrt{2}}{9}$ (D) $\frac{\pi}{2}$

Ans. C

Sol. -

Q14. If $\theta = \cot^{-1} \sqrt{\cos x} - \tan^{-1} \sqrt{\cos x}$, then $\sin \theta =$

- (A) $\tan \frac{1}{2} x$ (B) $\tan^2 (x/2)$ (C) $\frac{1}{2} \tan^{-1} (x/2)$ (D) None of these

Ans. B

Sol. Given that $\cot^{-1} (\sqrt{\cos \alpha}) - \tan^{-1} (\sqrt{\cos \alpha}) = x$ (i)

We know that, $\cot^{-1} (\sqrt{\cos \alpha}) + \tan^{-1} (\sqrt{\cos \alpha}) = \frac{\pi}{2}$

(ii)

On adding equation (i) and (ii), we get

$$2 \cot^{-1} (\sqrt{\cos \alpha}) = \frac{\pi}{2} + x \Rightarrow (\sqrt{\cos \alpha}) = \cot \left(\frac{\pi}{4} + \frac{x}{2} \right) \Rightarrow (\sqrt{\cos \alpha}) = \frac{\cot \frac{x}{2} - 1}{1 + \cot \frac{x}{2}}$$
$$\Rightarrow (\sqrt{\cos \alpha}) = \frac{\cos \frac{x}{2} - \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}} \Rightarrow \cos \alpha = \frac{1 - \sin x}{1 + \sin x} \Rightarrow \frac{1 - \tan^2 \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} = \frac{1 - \sin x}{1 + \sin x}$$

Applying componendo rule, we get $\sin x = \tan^2 \left(\frac{\alpha}{2} \right)$

Q15. If a, b, c be positive real numbers and the value of

$\theta = \tan^{-1} \sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1} \sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1} \sqrt{\frac{c(a+b+c)}{ab}}$ then $\tan \theta$ is equal to -

- (A) 0 (B) 1 (C) $\frac{a+b+c}{abc}$ (D) None of these

Ans. A

Sol. Let $x = \sqrt{\frac{a(a+b+c)}{bc}}$, $y = \sqrt{\frac{b(a+b+c)}{ac}}$, $z = \sqrt{\frac{c(a+b+c)}{ab}}$, $x, y, z > 0$

$$\Rightarrow \theta = \tan^{-1} x + \tan^{-1} y + \tan^{-1} z$$

$$\text{Now } x + y + z = \sqrt{\frac{a(a+b+c)}{bc}} + \sqrt{\frac{b(a+b+c)}{ac}} + \sqrt{\frac{c(a+b+c)}{ab}} = \sqrt{\frac{c(a+b+c)^{3/2}}{ab}}$$

$$\text{and } xyz = \frac{(a+b+c)^{3/2}}{\sqrt{abc}} \Rightarrow x + y + z = xyz \Rightarrow \tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$$

Q16. If $3 \cos^{-1}(x^2 - 7x + 25/2) = \pi$, then $x =$

- (A) Only 3 (B) Only 4 (C) 3 or 4 (D) None of these

Ans. C

Sol. -

Q17. If $\tan(x+y) = 33$ and $x = \tan^{-1} 3$, then y will be -

- (A) 0.3 (B) $\tan^{-1}(1.3)$ (C) $\tan^{-1}(0.3)$ (D) $\tan^{-1}\left(\frac{1}{18}\right)$

Ans. C

Sol. -

Q18. If $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi}{8}$, then x equals-

- (A) -1 (B) 1 (C) 0 (D) इनमें से कोई नहीं

Ans. A

Sol. -

Q19. If $\tan^{-1} \frac{1}{a-1} = \tan^{-1} \frac{1}{x} + \tan^{-1} \frac{1}{a^2-x+1}$, then x is -

- (A) $\frac{a}{2}$ (B) a^3 (C) $a^2 - a + 1$ (D) $a^2 + a - 1$

Ans. C

Sol. -

Q20. The trigonometric equation, $\sin^{-1} x = 2 \sin^{-1} a$ has a solution for :

- (A) all real values of a (B) $|a| < \frac{1}{2}$ (C) $|a| \leq \frac{1}{\sqrt{2}}$ (D) $\frac{1}{2} < |a| < \frac{1}{\sqrt{2}}$

Ans. C

Sol. $\sin^{-1} x = 2\sin^{-1} a$

$$\frac{\pi}{2} - \leq \sin^{-1} x \leq \frac{\pi}{2}$$

$$-\frac{\pi}{2} \leq 2\sin^{-1} a \leq \frac{\pi}{2} \Rightarrow -\frac{\pi}{4} \leq \sin^{-1} a \leq \frac{\pi}{4}$$

$$-\frac{1}{\sqrt{2}} \leq a \leq \frac{1}{\sqrt{2}}$$

Q21. If $\cos^{-1} x - \cos^{-1} \left(\frac{1}{2} \right) = \alpha$, then $4x^2 - 4xy \cos \alpha + y^2$ is equal to :

(A) 4 (B) $2 \sin 2\alpha$ (C) $-4 \sin^2 \alpha$ (D) $4 \sin^2 \alpha$

Ans. D

Sol. $\cos^{-1} \left[\frac{xy}{2} + \sqrt{1-x^2} \sqrt{1-\frac{y^2}{4}} \right] = \alpha$

$$\left(\frac{xy}{2} - \cos \alpha \right)^2 = (1-x^2) \left(1 - \frac{y^2}{4} \right)$$

$$\frac{x^2 y^2}{4} - \frac{2xy}{2} \cos^2 \alpha + \cos^2 \alpha = 1 - x^2 - \frac{y^2}{4} + \frac{x^2 y^2}{4}$$

$$x^2 + \frac{y^2}{4} - xy \cos^2 \alpha + \sin^2 \alpha$$

$$4x^2 + y^2 - 4xy \cos^2 \alpha = 4 \sin^2 \alpha$$

Q22. The sum of the infinite terms of the series $\cot^{-1} \left(1^2 + \frac{3}{4} \right) + \cot^{-1} \left(1^2 + \frac{3}{4} \right) + \cot^{-1} \left(3^2 + \frac{3}{4} \right) + \dots$ is equal to :

(A) $\tan^{-1} (1)$ (B) $\tan^{-1} (2)$ (C) $\tan^{-1} (3)$ (D) $\tan^{-1} (4)$

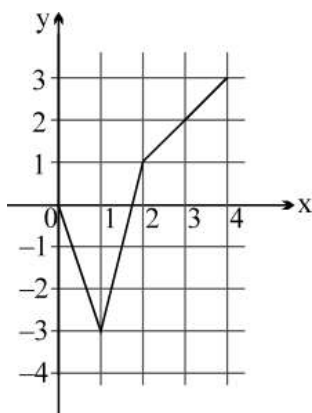
Ans. B

Sol. $T_n = \cot^{-1} \left(n^2 + \frac{3}{4} \right) = \tan^{-1} \left(\frac{1}{n^2 + \left(\frac{3}{4} \right)} \right) = \tan^{-1} \left(\frac{1}{1 + n^2 - \left(\frac{1}{4} \right)} \right)$

$$= \tan^{-1} \left(\frac{1}{1 + \left(n - \frac{1}{2} \right) \left(n + \frac{1}{2} \right)} \right) = \tan^{-1} \left(\frac{\left(n + \frac{1}{2} \right) - \left(n - \frac{1}{2} \right)}{1 + \left(n + \frac{1}{2} \right) \left(n - \frac{1}{2} \right)} \right)$$

Exercise J02

- Q1. Given is a partial graph of an even periodic function f whose period is 8. If $[]$ denotes greatest integer function then find the value of the expression.



$$f(-3) + 2 | f(-1) | + \left[f\left(\frac{7}{8}\right) \right] + f(0) + \arccos + f(-7) + f(20)$$

Ans. 5

Sol. -

- Q2. Find the domain of definition the following functions.
(Read the symbols $[]$ and $\{ \}$ as greatest integers and fractional part functions respectively.)

$$(i) f(x) = \arccos \cos \frac{2x}{1+x} \quad (ii) f(x) = \sqrt{\cos(\sin x)} + \sin^{-1} \frac{1+x^2}{2x}$$

$$(iii) f(x) = \sin^{-1} \left(\frac{x-3}{2} \right) - \log_{10}(4-x) \quad (iv) f(x) = \sin^{-1}(2x+x^2)$$

$$(v) f(x) = \frac{\sqrt{1-\sin x}}{\log_5(1-4x^2)} + \cos^{-1}(1-\{x\})$$

$$(vi) f(x) = \sqrt{3-x} + \cos^{-1} \left(\frac{3-2x}{5} \right) + \log_6(2|x|-3) + \sin^{-1}(\log_2 x)$$

$$(vii) f(x) = \log_{10}(1 - \log_7(x^2 - 5x + 13)) + \cos^{-1} \left(\frac{3}{2 + \sin \frac{9\pi x}{2}} \right)$$

$$(viii) f(x) = e^{\sin^{-1} \left(\frac{x}{2} \right)} + \tan^{-1} \left[\frac{x}{2} - 1 \right] + \ln \left(\sqrt{x - [x]} \right)$$

Ans. (i) $-1/3 \leq x \leq 1$ (ii) $\{1, -1\}$ (iii) $1 \leq x < 4$ (iv) $[-(1 + \sqrt{2}), (\sqrt{2}, -1)]$

(v) $x \in (-1/2, 1/2), x \neq 0$ (vi) $(3/2, 2]$ (vii) $\{7/3, 25/9\}$

(viii) $(-2, 2) - \{-1, 0, 1\}$ (ix) $\{x | x = 2n\pi + \frac{\pi}{6}, n \in \mathbb{I}\}$

Sol. -

- Q3. Identify the pair(s) of functions which are identical.

$$(a) y = \tan(\cos^{-1} x); y = \frac{\sqrt{1-x^2}}{x}$$

$$(b) y = \tan(\cot^{-1} x); y = \frac{1}{x}$$

$$(c) y = \sin(\arctan x); y = \frac{x}{\sqrt{1+x^2}}$$

$$(d) y = \cos(\arctan x); y = \sin(\arccot x)$$

Ans. (a), (b), (c) and (d) all are identical.

Sol. -

Q4. Find the domain and range of the following functions .

(Read the symbols $[]$ and $\{\}$ as greatest integers and fractional part functions respectively.)

$$(i) f(x) = \cot^{-1}(2x - x^2) \quad (ii) f(x) = \sec^{-1}(\log_3 \tan x + \log_{\tan x} 3)$$

$$(iii) f(x) = \cos^{-1}\left(\frac{\sqrt{2x^2+1}}{x^2+1}\right) \quad (iv) f(x) = \tan^{-1}\left(\log_{\frac{4}{5}}(5x^2 - 8x + 4)\right)$$

Ans. (i) $D : x \in \mathbb{R} \quad R : [\pi/4, \pi)$

$$(ii) D : x \in \left(n\pi, n\pi + \frac{\pi}{2}\right) - \left\{x \mid x = n\pi + \frac{\pi}{4}\right\} n \in I; \quad R : \left[\frac{\pi}{3}, \frac{2\pi}{3}\right] - \left[\frac{\pi}{2}\right]$$

$$(iii) D : x \in \mathbb{R} \quad R : \left[0, \frac{\pi}{2}\right) \quad (iv) D : x \in \mathbb{R} \quad R : \left(-\frac{\pi}{2}, \frac{\pi}{4}\right]$$

Sol. -

Q5. Let $f(x) = \cot^{-1}(x^2 + 4x + \alpha^2 - \alpha)$ be a function defined $\mathbb{R} \rightarrow (0, \pi/2]$ then find the complete set of real values of α for which $f(x)$ is onto.

Ans. $\frac{1 \pm \sqrt{17}}{2}$

Sol. $f(x) = \cot^{-1}(x^2 + 4x + \alpha^2 - \alpha)$

$$\Rightarrow x^2 + 4x + \alpha^2 - \alpha \geq 0$$

Now $f(x)$ has to be onto

$$\Rightarrow D = 0$$

$$\Rightarrow 16 - 4(\alpha^2 - \alpha) = 0$$

$$\Rightarrow \alpha = \frac{1 \pm \sqrt{17}}{2}$$

Q6. If the range of the function $\cos^{-1}|1 - x^2|$ is $\left[a, \frac{\pi}{b}\right]$ then find the value of b

Ans. 2

Sol. -

Q7. (a) Find the following

$$(i) \tan \left[\cos^{-1} \frac{1}{2} + \tan^{-1} \left(\frac{-1}{\sqrt{3}} \right) \right]$$

$$(ii) \cos^{-1} \left(\cos \frac{7\pi}{6} \right)$$

$$(iii) \cos \left(\tan^{-1} \frac{3}{4} \right)$$

$$(iv) \tan \left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} \right)$$

(b) Find the following :

$$(i) \sin \left[\frac{\pi}{2} - \sin^{-1} \left(\frac{-\sqrt{3}}{2} \right) \right]$$

$$(ii) \cos \left[\cos^{-1} \left(\frac{-\sqrt{3}}{2} \right) + \frac{\pi}{6} \right]$$

$$(iii) \tan^{-1} \left(\tan \frac{3\pi}{4} \right)$$

$$(iv) \sin \left(\frac{1}{4} \arcsin \frac{\sqrt{63}}{8} \right)$$

Ans. -

Sol. -

Q8. Show that :

$$\sin^{-1} \left(\sin \frac{33\pi}{7} \right) + \cos^{-1} \left(\cos \frac{46\pi}{7} \right) + \tan^{-1} \left(-\tan \frac{13\pi}{8} \right) + \cot^{-1} \left(\cot \left(-\frac{19\pi}{8} \right) \right) = \frac{13\pi}{7}$$

Ans. -

Sol. -

Q9. Find the simplest value of

$$(a) f(x) = \arccos x + \arccos \left(\frac{x}{2} + \frac{1}{2} \sqrt{3-3x^2} \right), \quad x \in \left(\frac{1}{2}, 1 \right)$$

$$(b) f(x) = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right), \quad x \in \mathbb{R} - \{0\}$$

Ans. -

Sol. -

Q10. Evaluate : $\left| \tan^{-1} \left(\frac{x-y}{1+xy} \right) + \tan^{-1} \left(\frac{y-z}{1+yz} \right) + \tan^{-1} \left(\frac{z-x}{1+zx} \right) \right|$

(i) when x, y, z all are of same sign.

(ii) When not all three are of the same sign and (one positive $\times$ one negative) < -1 and

(one positive $\times$ one negative) > -1 .

Ans. (i) 0 (ii) π

Sol. -

Q11. Prove that : $\sin \cot^{-1} \tan \cos^{-1} x = \sin \operatorname{cosec}^{-1} \cot \tan^{-1} x = x$ where $x \in (0, 1]$

Ans. -

Sol. -

Q12. Prove the identities.

$$(a) \sin^{-1} \cos (\sin^{-1} x) + \cos^{-1} \sin (\cos^{-1} x) = \frac{\pi}{2}, \quad |x| \leq 1$$

$$(b) 2 \tan^{-1} (\operatorname{cosec} \tan^{-1} x - \tan \cot^{-1} x) = \tan^{-1} x \quad (x \neq 0)$$

$$(c) \tan (\tan^{-1} x + \tan^{-1} y + \tan^{-1} z) = \cot (\cot^{-1} x + \cot^{-1} y + \cot^{-1} z)$$

Ans. -

Sol. -

Q13. Let $\alpha = \sin^{-1} \left(\frac{36}{85} \right)$, $\beta = \cos^{-1} \left(\frac{4}{5} \right)$ and $\gamma = \tan^{-1} \left(\frac{8}{15} \right)$, find $(\alpha + \beta + \gamma)$ and hence prove that

$$(i) \sum \cot \alpha = \prod \cot \alpha, \quad (ii) \sum \tan \alpha \cdot \tan \beta = 1$$

Ans. $\pi/2$

Sol. -

Q14. Prove that: (a) $2 \cos^{-1} \frac{3}{\sqrt{13}} + \cot^{-1} \frac{16}{63} + \frac{1}{2} \cos^{-1} \frac{7}{25} = \pi$

(b) $\cot^{-1} \left(\frac{5}{13} \right) + \cos^{-1} \left(-\frac{7}{25} \right) + \sin^{-1} \frac{36}{325} = \pi$ (c) $\arccos \sqrt{\frac{2}{3}} - \arccos =$

Ans. -

Sol. -

Q15. If α and β are the roots of the equation $x^2 + 5x - 49 = 0$ then find the value of $\cot(\cot^{-1} \alpha + \cot^{-1} \beta)$.

Ans. -

Sol. -

Q16. If $a > b > c > 0$ then find the value of : $\cot^{-1} \left(\frac{ab+1}{a-b} \right) + \cot^{-1} \left(\frac{bc+1}{b-c} \right) + \cot^{-1} \left(\frac{ca+1}{c-a} \right)$.

Ans. π

Sol. -

Q17. Prove that : $\tan^{-1} n + \cot^{-1} (n+1) = \tan^{-1} (n^2 + n + 1)$.

Ans. -

Sol. -

Q18. In a triangle ABC if $\angle A = 90^\circ$, then prove that , $\tan^{-1} \frac{b}{c+a} + \tan^{-1} \frac{c}{a+b} = \frac{\pi}{4}$

Ans. -

Sol. -

Q19. Solve for x and y : $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$ and $\cos^{-1} x - \cos^{-1} y = \frac{\pi}{3}$

Ans. $x = \frac{1}{2}, y = 1$

Sol. -

Q20. Find all possible value of 'p' and 'q' for which $\cos^{-1} \sqrt{p} + \cos^{-1} \sqrt{1-p} + \cos^{-1} \sqrt{1-q} = \frac{3\pi}{4}$

Ans. $0 \leq p \leq 1$ and $q = \frac{1}{2}$

Sol. -

Q21. Solve the following equations:

$$(a) \tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$$

$$(b) \tan^{-1} \frac{x-1}{x+1} + \tan^{-1} \frac{2x-1}{2x+1} = \tan^{-1} \frac{23}{36}$$

$$(c) \cos^{-1} \frac{x^2-1}{x^2+1} + \tan^{-1} \frac{2x}{x^2-1} = \frac{2\pi}{36}$$

Ans. (a) $x = 0, \frac{1}{2}, -\frac{1}{2}$; (b) $x = \frac{4}{3}$; (c) $x = 2 - \sqrt{3}$ or $\sqrt{3}$

Sol. -

Q22. $\tan^{-1} \frac{1}{x^2 + x + 1} + \tan^{-1} \frac{1}{x^2 + 3x + 3} + \tan^{-1} \frac{1}{x^2 + 5x + 7} + \dots$
to n terms, where $x > 0$.

Ans. $\tan^{-1}(x+n) - \tan^{-1} x$

Sol. $\tan^{-1} \left(\frac{1}{x^2 + x + 1} \right) + \tan^{-1} \left(\frac{1}{x^2 + 3x + 3} \right) + \tan^{-1} \left(\frac{1}{x^2 + 5x + 7} \right) + \dots$

$$1 + 3 + 5 + \dots = n^2$$

$$S = 1 + 3 + 5 + \dots + T_n$$

$$S = 1 + 3 + 5 + \dots + T_{n-1} + T_n$$

$$S - S = 1 + 2 + 2 + \dots - T_n$$

$$T_n = 1 + \frac{n-1}{2} [2 \times 2 + (n-1)2]$$

$$T_n = n^2 - n + 1$$

$$T_n = \tan^{-1} \left[\frac{1}{x^2 + (2n-1)x + (n^2 - n + 1)} \right]$$

$$= \tan^{-1} \left[\frac{1}{1 + x^2 + nx + (n-1)x + n(x-1)} \right]$$

$$= \tan^{-1} \left[\frac{1}{1 + x(n+x) + (n-1)(n+x)} \right]$$

$$= \tan^{-1} \left[\frac{1}{1 + (n+x)(n+x-1)} \right]$$

$$= \tan^{-1} \left[\frac{(n+x) - (x+n-1)}{1 + (n+x)(n+x-1)} \right]$$

$$T_n = \tan^{-1}(n+x) - \tan^{-1}(n+x-1)$$

$$T_1 = \tan^{-1}(1+x) - \tan^{-1}(x)$$

$$T_2 = \tan^{-1} (2 + x) - \tan^{-1} (1 + x)$$

$$T_3 = \tan^{-1} (3 + x) - \tan^{-1} (2 + x)$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$T_n = \tan^{-1} (n + x) - \tan^{-1} (n + x - 1)$$

$$S_n = \tan^{-1} (n + x) - \tan^{-1} x$$

$$\text{Q23.} \quad \sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{\sqrt{2} - 1}{\sqrt{6}} + \dots + \sin^{-1} \frac{\sqrt{n} - \sqrt{n-1}}{\sqrt{n(n+1)}} + \dots$$

$$\text{Ans.} \quad \pi/2$$

$$\text{Sol.} \quad \sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{\sqrt{2} - 1}{\sqrt{6}} + \dots + \sin^{-1} \frac{\sqrt{n} - \sqrt{n-1}}{\sqrt{n(n+1)}} + \dots$$

$$T_n = \sin^{-1} \left(\frac{\sqrt{n} - \sqrt{n-1}}{\sqrt{n}\sqrt{n+1}} \right)$$

$$n^2 + x - n - n + 1 - 2\sqrt{n(n-1)}$$

$$n^2 - n + 1 - 2\sqrt{n(n-1)}$$

$$1 + n(n-1) - 2\sqrt{n(n-1)}$$

$$(n^2 + n) - \left(n + n - 1 - 2\sqrt{n(n-1)} \right)$$

$$n^2 + n - n - n + 1 + 2\sqrt{n(n-1)}$$

$$\sqrt{n^2 - n + 1 + 2\sqrt{n(n-1)}}$$

$$\left(n\sqrt{n-1} \right)^2 + 1 + n2\sqrt{n(n-1)}$$

$$\sqrt{n(n-1)} + 1$$

$$S_n = \tan^{-1} \sqrt{n} - 0$$

$$S_n = \tan^{-1} + \infty$$

$$= \frac{\pi}{2}$$

Paragraph

Consider the function $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ and $h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$.

\n

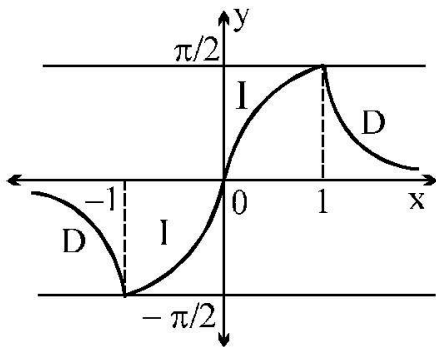
Q24. If $x \in (-1, 1)$, then the solution of the equation $f(x) + g(x) + h(x) = \frac{\pi}{2}$, is equal to

- (A) $\sqrt{2} - 1$ (B) $2 - \sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{2}$

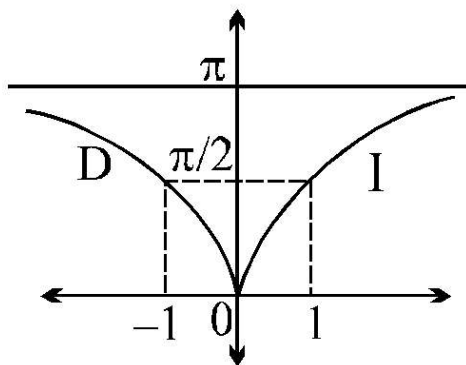
Ans. B

Sol. Starting with $x = \tan \theta$

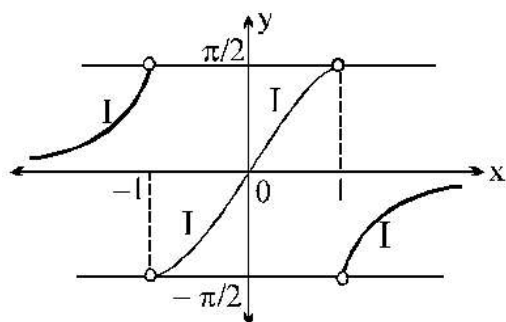
$$f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & -1 \leq x \leq 1 \\ \pi - 2\tan^{-1}x & \text{if } x \geq 1 \\ -\pi - 2\tan^{-1}x & x \leq -1 \end{cases}$$



$$g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & x \geq 0 \\ -2\tan^{-1}x & x < 0 \end{cases}$$



$$h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \begin{cases} \pi + 2\tan^{-1}x & x < -1 \\ 2\tan^{-1}x & -1 < x < 1 \\ 2\tan^{-1}x - \pi & x > 1 \end{cases}$$



in $(0, 1)$, $f(x) + g(x) + h(x) =$

$$2 \tan^{-1} x + 2 \tan^{-1} x + 2 \tan^{-1} x = \frac{\pi}{12} \Rightarrow \tan^{-1} x = \frac{\pi}{12} \Rightarrow x = \tan \frac{\pi}{12} = 2 - \sqrt{3}$$

Paragraph

Consider the function $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ and $h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$.

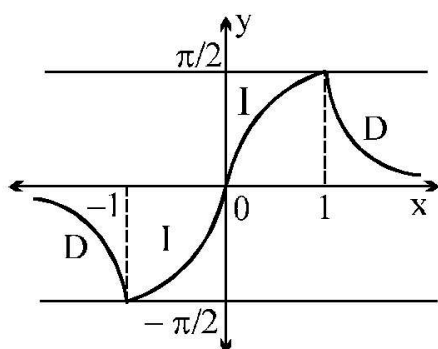
Q25. $f(2) + g(2) + h(2)$ has the value equal to

(A) $\cot^{-1}\left(\frac{-4}{3}\right)$ (B) $\cot^{-1}\left(\frac{-3}{4}\right)$ (C) $\pi - \tan^{-1}\left(\frac{2}{3}\right)$ (D) $\cos^{-1}\left(\frac{-3}{4}\right)$

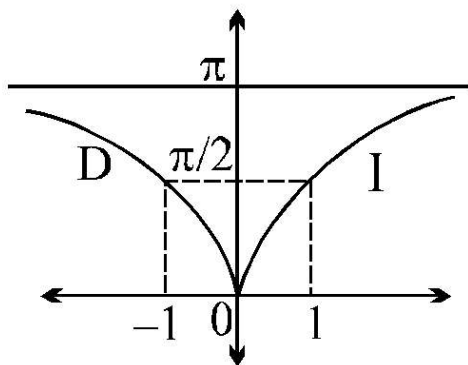
Ans. B

Sol. Starting with $x = \tan \theta$

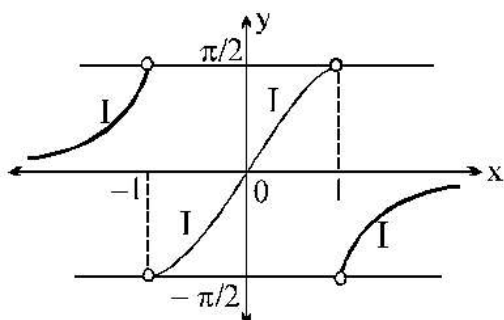
$$f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2\tan^{-1} x & -1 \leq x \leq 1 \\ \pi - 2\tan^{-1} x & \text{if } x \geq 1 \\ -\pi - 2\tan^{-1} x & x \leq -1 \end{cases}$$



$$g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1} x & x \geq 0 \\ -2\tan^{-1} x & x < 0 \end{cases}$$



$$h(x) = \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \begin{cases} \pi + 2 \tan^{-1} x & x < -1 \\ 2 \tan^{-1} x & -1 < x < 1 \\ 2 \tan^{-1} x - \pi & x > 1 \end{cases}$$



$$\begin{aligned} \sin^{-1} \left(\frac{4}{5} \right) + \cos^{-1} \left(\frac{-3}{5} \right) + \tan^{-1} \left(\frac{4}{-3} \right) &= \tan^{-1} \left(\frac{4}{3} \right) + \pi - \cos^{-1} \left(\frac{3}{5} \right) - \tan^{-1} \left(\frac{4}{3} \right) \\ &= \pi - \tan^{-1} \left(\frac{4}{3} \right) = -\cot^{-1} \left(\frac{4}{3} \right) = \cot^{-1} \left(\frac{-3}{4} \right) \text{ Ans.} \end{aligned}$$

Paragraph

$$\text{Given that } \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \begin{cases} 2 \tan^{-1} x & , \quad |x| < 1 \\ -\pi + 2 \tan^{-1} x & , \quad x > 1 \\ \pi + 2 \tan^{-1} x & , \quad x < -1 \end{cases}$$

$$\sin^{-1} \frac{2x}{1+x^2} = \begin{cases} 2 \tan^{-1} x & \text{is } |x| \leq 1 \\ \pi - 2 \tan^{-1} x & \text{is } x > 1 \\ -(\pi + 2 \tan^{-1} x) & \text{is } x < -1 \end{cases}$$

$$\text{and } \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \text{ for } -1 \leq x \leq 1$$

\n

Q26. $\sin^{-1} \left(\frac{4x}{x^2+4} \right) + 2 \tan^{-1} \left(-\frac{x}{2} \right)$ is independent from x then:

- (A) $x \in [-3, 4]$ (B) $x \in [-2, 2]$ (C) $x \in [-1, 1]$ (D) $x \in [1, \infty]$

Ans. B

Sol. -

Paragraph

$$\text{Given that } \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \begin{cases} 2 \tan^{-1} x & , \quad |x| < 1 \\ -\pi + 2 \tan^{-1} x & , \quad x > 1 \\ \pi + 2 \tan^{-1} x & , \quad x < -1 \end{cases}$$

$$\sin^{-1} \frac{2x}{1+x^2} = \begin{cases} 2 \tan^{-1} x & \text{is } |x| \leq 1 \\ \pi - 2 \tan^{-1} x & \text{is } x > 1 \\ -(\pi + 2 \tan^{-1} x) & \text{is } x < -1 \end{cases}$$

$$\text{and } \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \text{ for } -1 \leq x \leq 1$$

\n

$$\text{Q27. If } \cos^{-1} \frac{6x}{1+9x^2} = -\frac{\pi}{2} + 2 \tan^{-1} 3x, \text{ then } x \in$$

- (A) $\left(\frac{1}{3}, \infty\right)$ (B) $(-1, \infty)$ (C) $(-\infty, -1)$ (D) None of these

Ans. A

Sol. -

Paragraph

$$\text{Given that } \tan^{-1} \left(\frac{2x}{1-x^2} \right) = \begin{cases} 2 \tan^{-1} x & , \quad |x| < 1 \\ -\pi + 2 \tan^{-1} x & , \quad x > 1 \\ \pi + 2 \tan^{-1} x & , \quad x < -1 \end{cases}$$

$$\sin^{-1} \frac{2x}{1+x^2} = \begin{cases} 2 \tan^{-1} x & \text{is } |x| \leq 1 \\ \pi - 2 \tan^{-1} x & \text{is } x > 1 \\ -(\pi + 2 \tan^{-1} x) & \text{is } x < -1 \end{cases}$$

$$\text{and } \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \text{ for } -1 \leq x \leq 1$$

\n

$$\text{Q28. If } (x-1)(x^2+1) > 0, \text{ then } \sin \left(\frac{1}{2} \tan^{-1} \frac{2x}{1-x^2} - \tan^{-1} x \right) =$$

- (A) 1 (B) $\frac{1}{\sqrt{2}}$ (C) -1 (D) None of these

Ans. C

Sol. -

Q29. In the following functions defined from $[-1,1]$ to $[-1,1]$, then functions which are not bijective are :

- (A) $\sin(\sin^{-1} x)$ (B) $\frac{2}{\pi} \sin^{-1}(\sin x)$ (C) $(\operatorname{sgn} x) \ln e^x$ (D) $x^3 \operatorname{sgn} x$

Ans. B, C, D

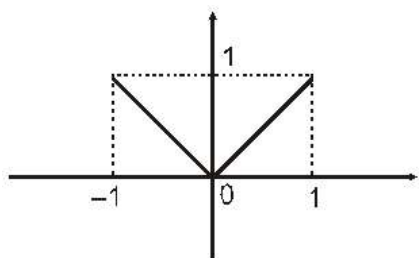
Sol. $f : [-1, 1] \rightarrow [-1, 1]$

(A) $f(x) = \sin(\sin^{-1} x) = x$, $x \in [-1, 1]$ Bijective $y \in [-1, 1]$

(B) $f(x) = \frac{2}{\pi} \sin^{-1}(\sin x) = \frac{2x}{\pi}$, $x \in [-1, 1]$

Not bijective $y \in \left[\frac{-2}{\pi}, \frac{2}{\pi} \right]$

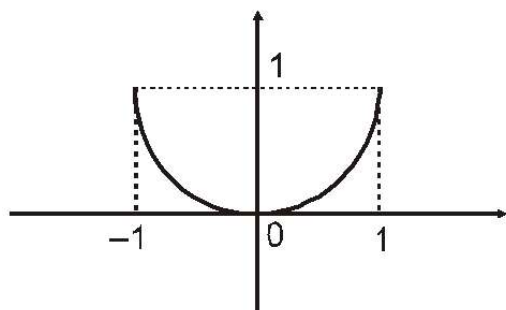
(C) $f(x) = (\operatorname{sgn} x)(\ln e^x)$



$$= \begin{cases} x & , \quad x \in (0, 1] \\ 0 & , \quad x = 0 \\ -x & , \quad x \in [-1, 0) \end{cases}$$

Not bijective

(D) $f(x) = x^3 \operatorname{sgn} x$



$$\begin{cases} x^3 & , \quad x > 0 \\ 0 & , \quad x = 0 \\ -x^3 & , \quad x < 0 \end{cases}$$

Not bijective

Q30. If α, β, γ are three angles given by $\alpha = 2 \tan^{-1}(\sqrt{2} - 1)$, $\beta = 3 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ + $\sin^{-1}\left(-\frac{1}{2}\right)$ and $\gamma = \cos^{-1}\frac{1}{3}$, then :

(A) $\alpha > \beta$ (B) $\beta > \gamma$ (C) $\alpha < \gamma$ (D) $\beta < \gamma$

Ans. B, C

Sol. $\alpha = 2 \tan^{-1} \left(\sqrt{2} - 1 \right) = 2 \times 15 = 30 = \frac{\pi}{6} = \frac{2\pi}{12}$

$$\beta = 3 \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) + \sin^{-1} \left(-\frac{1}{2} \right)$$

$$= 3 \times \frac{\pi}{4} - \frac{\pi}{6} = \frac{7\pi}{12}$$

$$\gamma = \cos^{-1} \left(\frac{1}{3} \right)$$

$$\frac{\pi}{3} < \gamma < \frac{\pi}{2}$$

$$\frac{4\pi}{12} < \gamma < \frac{6\pi}{12}$$

Q31. Which of the following are identical functions?

(where $[x]$ denotes greatest integer less than or equal x , $\{x\}$ denotes fractional part of x and $\operatorname{sgn} x$ denotes signum function of x respectively.)

(A) $f(x) = \operatorname{sgn}(|x| + 1)$ (B) $g(x) = \sin^2(\ln x) + \cos^2(\ln x)$

(C) $h(x) = \frac{2}{\pi}(\sin^{-1}\{x\} + \cos^{-1}\{x\})$ (D) $k(x) = \sec^2[\{x\}] - \tan^2\{[x]\}$

Ans. C, A, D

Sol. (A) $\operatorname{sgn}(|x| + 1) = 1 \forall x \in \mathbb{R}$.

(B) $\sin^2(\ln x) + \cos^2(\ln x) = 1 \forall x \in \mathbb{R}^+$

(C) $\frac{2}{\pi}(\sin^{-1}\{1\} + \cos^{-1}\{x\}) = 1 \forall x \in \mathbb{R}$.

(D) $\sec^2[\{x\}] - \tan^2\{[x]\} = \sec^2 0^\circ - \tan^2 0^\circ = 1 \forall x \in \mathbb{R}$.

Q32. The value of a for which $a x + \sec^{-1} \sqrt{2x^2 - x^4} + \cos^{-1} \sqrt{2x^2 - x^4} = 0$

(A) $\frac{\pi}{2}$ (B) $-\frac{\pi}{2}$ (C) $\frac{2}{\pi}$ (D) $-\frac{2}{\pi}$

Ans. A, B

Sol. for existence of $\sec^{-1}(\sqrt{2x^2 - x^4})$ or $\operatorname{cosec}^{-1}(\sqrt{2x^2 - x^4})$

$$2x^2 - x^4 \leq 1$$

i.e. $(x^2 - 1)^2 \leq 0$ hence $x^2 = 1 \Rightarrow x = \pm 1$

if $x = 1$, $a + \frac{\pi}{2} = 0$;

if $x = -1$, $-a + \frac{\pi}{2} = 0 \Rightarrow a = \pm \frac{\pi}{2}$ Ans.

Q33. Let function $f(x)$ be defined as $f(x) = |\sin^{-1}x| + \cos^{-1}\left(\frac{1}{x}\right)$. Then which of the following is/are True?

(A) $f(x)$ is injective in its domain. (B) $f(x)$ is many-one in its domain.

(C) Range of f is a singleton set.

(D) $\operatorname{sgn} f(x) = 1$ where $\operatorname{sgn} x$ denotes signum function of x .

Ans. A, D

Sol. $f(x) = |\sin^{-1}x| + \cos^{-1}\left(\frac{1}{x}\right)$

Domain of $f(x)$ is $\{-1, 1\}$

$$f(1) = \frac{\pi}{2}, f(-1) = \frac{3\pi}{2}$$

So function $f(x)$ is injective

$$\operatorname{sgn}(f(x)) = 1 \quad (f(x) > 0)$$

$$\text{Range of } f(x) = \left\{ \frac{\pi}{2}, \frac{3\pi}{2} \right\}$$

Q34. Which of the following statement(s) is/are **true**?

(A) Domain of $y = \cos^{-1}(e^x)$ is same as range of $y = -\sqrt{-x}$.

(B) Number of elements common in the range of function $y = \tan^{-1}(\operatorname{sgn} x)$ and $y = \cot^{-1}(\operatorname{sgn} x)$ is only 1 (where $\operatorname{sgn} x$ denotes signum function of x .)

(C) The function $y = \operatorname{sgn}(\cot^{-1}x)$ and $y = 1$ are identical functions.

(D) Number of integers in the solution set of $1 < \log_2(\tan^{-1}x) < 2$ is 4.

Ans. A, B, C

Sol. (A) For domain of $y = \cos^{-1}(e^x)$, $0 < e^x \leq 1 \Rightarrow x \leq 0$ or $x \in (-\infty, 0)$

For range of $y = -\sqrt{-x}$, $-y = \sqrt{-x}$

$$\sqrt{-x} \geq 0 \Rightarrow y \geq 0 \Rightarrow y \in (-\infty, 0)$$

$\therefore$ Domain of $y = \cos^{-1}(e^x)$ is identical with range of $y = -\sqrt{-x}$

$$(B) y = \tan^{-1}(\operatorname{sgn} x) = \begin{cases} \tan^{-1}(-1), & \forall x < 0 \\ \tan^{-1} 0, & \text{for } x = 0 \\ \tan^{-1}(1), & \forall x > 0 \end{cases} \Rightarrow \text{Range of } y = \tan^{-1}(\operatorname{sgn} x) \text{ is } \left\{ \frac{-\pi}{4}, 0, \frac{\pi}{4} \right\}$$

$$\text{Also, } y = \cot^{-1}(\operatorname{sgn} x) = -\tan^{-1}(\operatorname{sgn} x) \Rightarrow \text{Range of } y = \cot^{-1}(\operatorname{sgn} x) \text{ is } \left\{ \frac{3\pi}{4}, \frac{\pi}{2}, \frac{\pi}{4} \right\}$$

$\therefore$ Number of elements common in the range of $y = \tan^{-1}(\operatorname{sgn} x)$ and $y = \cot^{-1} \operatorname{sgn} x$ is only one i.e. $\frac{\pi}{4}$.

$$(C) y = \operatorname{sgn}(\cot^{-1} x) = 1 \quad \forall x \in \mathbb{R}$$

$\therefore y = \operatorname{sgn}(\cot^{-1} x)$ and $y = 1$ are identical functions.

$$(D) 1 < \log_2(\tan^{-1} x) < 2 \Rightarrow 2^1 < \tan^{-1} x < 2^2 \Rightarrow \text{Not possible.} \\ \left(\text{As } \frac{-\pi}{2} < \tan^{-1} x < \frac{\pi}{2} < 2 \right)$$

Q35. $2\cot^{-1} 7 + \cos^{-1} \frac{3}{5}$ is equal to-

$$(A) \cot^{-1} \frac{44}{117} \quad (B) \operatorname{cosec}^{-1} \frac{125}{117} \quad (C) \tan^{-1} \frac{4}{117} \quad (D) \cos^{-1} \frac{44}{125}$$

Ans. D, B, A

Sol. -

Q36. If $\tan^2(\tan^{-1} x) = \operatorname{cosec}^{-1}(\operatorname{cosec} 7) + \sec(\sec^{-1} 8) + \cot^{-1}(\cot(-6))$, then $\tan^{-1}(\tan(18 + x^2))$:-

(A) is always positive (B) is always negative (C) is irrational (D) is rational

Ans. B, C

Sol. -

Q37. Let $\tan^{-1}\left(\tan \frac{5\pi}{4}\right) = \alpha$, $\tan^{-1}\left(-\tan \frac{2\pi}{3}\right) = \beta$. Then

$$(A) \alpha > \beta \quad (B) 4\alpha - 3\beta = 0 \quad (C) \alpha + \beta = \frac{7\pi}{12} \quad (D) \alpha + \beta = \pi$$

Ans. B, C

Sol. -

Q38. $2 \tan(\tan^{-1}(x) + \tan^{-1}(x^3))$ where $x \in \mathbb{R} - \{-1, 1\}$ is equal to

$$(A) \frac{2x}{1-x^2} \quad (B) \tan(2 \tan^{-1} x) \quad (C) \tan(\cot^{-1}(-x) - \cot^{-1}(x)) \quad (D) \tan(2 \cot^{-1} x)$$

Ans. A, B, C

Sol. $2 \tan(\tan^{-1}(x) + \tan^{-1}(x^3))$

$$\begin{aligned}
&\Rightarrow 2 \tan \left(\tan^{-1} \left(\frac{x + x^3}{1 - x^4} \right) \right) \\
&= 2 \tan \left[\tan^{-1} \left(\frac{x(x^2 + 1)}{(1 - x^2)(1 + x^2)} \right) \right] \\
&= 2 \tan \left[\tan^{-1} \left(\frac{x}{1 - x^2} \right) \right] \\
&= \frac{2x}{1 - x^2}
\end{aligned}$$

$$(D) \quad \tan(2 \cot^{-1} x) \quad \cot^{-1} x = \theta; \theta \in (0, \pi)$$

$$\cot \theta = x$$

$$\tan \theta = \frac{1}{x}$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{\frac{2}{x}}{1 - \frac{1}{x^2}} = \frac{2x}{x^2 - 1}$$

$$(C) \quad \tan(\cot^{-1}(-x) - \cot^{-1} x)$$

$$\Rightarrow \tan(\pi - 2 \cot^{-1} x)$$

$$= -\tan(2 \cot^{-1} x)$$

$$\Rightarrow \cot^{-1} x = \theta$$

$$= \tan 2\theta$$

$$= \frac{-2 \tan \theta}{1 - \tan^2 \theta} = \frac{-\frac{2}{x}}{1 - \frac{1}{x^2}} = \frac{2x}{1 - x^2}$$

$$(B) \quad \tan(2 \tan^{-1} x)$$

$$\Rightarrow \tan^{-1} x = \theta$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2x}{1 - x^2}$$

Q39. For the equation $2x = \tan(2 \tan^{-1} a) + 2 \tan(\tan^{-1} a + \tan^{-1} a^3)$, which of the following is invalid ?

$$(A) \quad a^2 x + 2a = x \quad (B) \quad a^2 + 2ax + 1 = 0 \quad (C) \quad a \neq 0 \quad (D) \quad a \neq -1, 1$$

Ans. B, C

Sol. -

Q40. If integral pair (x, y) satisfy the equation $\arctan \frac{1}{x} + \arctan \frac{1}{y} = \arctan \frac{1}{10}$, then sum of x + y can be

- (A) 122 (B) 100 (C) 92 (D) -92

Ans. A

Sol. -

Q41. Column-I

Column-II

(A) Period of $\frac{1}{2} \left\{ \frac{|\sin x|}{\cos x} + \frac{|\cos x|}{\sin x} \right\}$

(P) 2

(B) Range of $\sin^{-1} \sqrt{\log_{[x]} \frac{|x|}{x}}$

(Q) 2π

(C) Total number of solution $x^2 - 4 - [x] = 0$

(R) 1

(D) Period of $f(x) = \sin^2 x + \cos x$

(S) $\{0\}$

[where [] denotes G.I.F]

(A) (A) Q (B) S (C) P (D) Q (B) (A) S (B) Q (C) P (D) Q

(C) (A) Q (B) S (C) Q (D) P (D) (A) R (B) P (C) S (D) Q

Ans. 5f11858eb58d37a192819b63

Sol. -

Q42. Column-I

Column-II

(A) $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$

(P) range is $[0, \pi]$

(B) $g(x) = \sin^{-1}|x| + 2 \tan^{-1}|x|$

(Q) is increasing $\forall x \in (0, 1)$

(C) $h(x) = 2 \sin^{-1} \left(\frac{2x}{1+x^2} \right) \quad x \in [0, 1]$

(R) period is 2π

(D) $k(x) = \cot(\cot^{-1}x)$

(S) is decreasing $\forall x \in (0, 1)$

(A) (A) R (B) PQ (C) PQ (D) QR (B) (A) PQR (B) PQ (C) PQ (D) Q

(C) (A) PQR (B) RQ (C) P (D) Q (D) (A) R (B) PQR (C) PQ (D) RQ

Ans. 5f1187ad90e4b0a1523c7256

Sol. -

Q43. If $S_n = \sum_{r=1}^n r!$ then for $n > 6$ $\left(\text{given } \sum_{r=1}^6 r! = 873 \right)$: (where [] denotes greatest integer function)

Column-I

Column-II

(A) $\sin^{-1} \left(\sin \left(S_n - 7 \left[\frac{S_n}{7} \right] \right) \right)$

(P) $5 - 2\pi$

(B) $\cos^{-1} \left(\cos \left(S_n - 7 \left[\frac{S_n}{7} \right] \right) \right)$

(Q) $2\pi - 5$

- (C) $\tan^{-1} \left(\tan \left(S_n - 7 \left[\frac{S_n}{7} \right] \right) \right)$ (R) $6 - 2\pi$
 (D) $\cot^{-1} \left(\cot \left(S_n - 7 \left[\frac{S_n}{7} \right] \right) \right)$ (S) $5 - \pi$
 (T) $\pi - 4$

- (A) (A) P (B) Q (C) P (D) S (B) (A) Q (B) P (C) P (D) S
 (C) (A) P (B) S (C) P (D) Q (D) (A) P (B) R (C) P (D) R

Ans. 5f118b97b58d37a192819b77

Sol. -

Q44. Let (x, y) be such that $\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}$

Match the statements in **Column I** with statements in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I

Column II

- (A) If $a = 1$ and $b = 0$, then (x, y) (P) lies on the circle $x^2 + y^2 = 1$
 (B) If $a = 1$ and $b = 1$, then (x, y) (Q) lies on $(x^2 - 1)(y^2 - 1) = 0$
 (C) If $a = 1$ and $b = 2$, then (x, y) (R) lies on $y = x$
 (D) If $a = 2$ and $b = 2$, then (x, y) (S) lies on $(4x^2 - 1)(y^2 - 1) = 0$

- (A) (A) P (B) R (C) R (D) S (B) (A) R (B) R (C) P (D) S
 (C) (A) P (B) Q (C) P (D) S (D) (A) P (B) Q (C) S (D) P

Ans. 5f118fb2b58d37a192819b8f

Sol. $\sin^{-1}(ax) + \cos^{-1}y + \cos^{-1}(bxy) = \frac{\pi}{2}$

(A) $a = 1, b = 0 \Rightarrow \sin^{-1}(x) + \cos^{-1}(y) + \cos^{-1}(0) = \frac{\pi}{2} \Rightarrow \sin^{-1}x + \cos^{-1}y = 0$

$\Rightarrow \cos^{-1}y = -\sin^{-1}x \Rightarrow \cos^{-1}y = \cos^{-1}\sqrt{1-x^2} \quad x^2 + y^2 = 1$

(B) $\sin^{-1}(x) + \cos^{-1}y + \cos^{-1}(xy) = \frac{\pi}{2} \Rightarrow \cos^{-1} + \cos^{-1}(xy) = \cos^{-1}x.$

$\Rightarrow \cos^{-1}\left(xy^2 - \sqrt{(1-y^2)(1-x^2y^2)}\right) = \cos^{-1}x. \Rightarrow xy^2 - \sqrt{(1-y^2)(1-x^2y^2)} = x$

$\Rightarrow 1 - x^2 - y^2 + x^2y^2 = 0 \Rightarrow (1-x^2)(1-y^2) = 0$

(C) $\sin^{-1}(x) + \cos^{-1}y + \cos^{-1}(2xy) = \frac{\pi}{2} \Rightarrow \cos^{-1}$

$\left(2xy^2 - \sqrt{(1-y^2)(1-4x^2y^2)}\right) = \cos^{-1}x.$

$\Rightarrow 2xy^2 - \sqrt{(1-y^2)(1-4x^2y^2)} = x \Rightarrow 2xy^2 - x = \sqrt{(1-y^2)(1-4x^2y^2)}$

$\Rightarrow 4x^2y^4 + x^2 - 4x^2y^2 = 1 - y^2 - 4x^2y^2 + 4x^2y^4 \Rightarrow \cos^{-1}$

$\left(2y^2x - \sqrt{(1-y^2)(1-4x^2y^2)}\right) = \cos^{-1}(2x)$

$$\Rightarrow 2y^2x - \sqrt{1 - y^2 - 4x^2y^2 + 4x^2y^4} = 2x$$

$$\Rightarrow (1 - 4x^2)(1 - y^2) = 0$$

$$\Rightarrow 1 - 4x^2 - y^2 + 4x^2y^2 = 0$$